To refine this argument to $\mathbb{Z}$ coefficients the problem of signs must be addressed. After analyzing the situation more closely, one sees that if M is orientable, it is possible to make consistent choices of orientations of all the cells of C and C^* so that the boundary maps in C agree with the coboundary maps in C^*, and therefore one gets $H_i(C;\mathbb{Z}) \approx H^{2-i}(C^*;\mathbb{Z})$, hence $H_i(M;\mathbb{Z}) \approx H^{2-i}(M;\mathbb{Z})$.

For manifolds of higher dimension the situation is entirely analogous. One would consider dual cell structures C and C^* on a closed n-manifold M, each i-cell of C being dual to a unique $(n-i)$-cell of C^* which it intersects in one point 'transversely.' For example on the 3-dimensional torus $S^1\times S^1\times S^1$ one could take the standard cell structure lifting to the decomposition of the universal cover $\mathbb{R}^3$ into cubes with vertices at the integer lattice points $\mathbb{Z}^3$, and then the dual cell structure is obtained by translating this by the vector $(1/2, 1/2, 1/2)$. Each edge in either cell structure then has a dual 2-cell which it pierces orthogonally, and each vertex lies in a dual 3-cell.

All the manifolds one commonly meets, for example all differentiable manifolds, have dually paired cell structures with the properties needed to carry out the proof of Poincaré duality we have just sketched. However, to construct these cell structures requires a certain amount of manifold theory. To avoid this, and to get a theorem that applies to all manifolds, we will take a completely different approach, using algebraic topology to replace the geometry of dual cell structures.

Orientations and Homology

Let us consider the question of how one might define orientability for manifolds. First there is the local question: What is an orientation of $\mathbb{R}^n$? Whatever an orientation of $\mathbb{R}^n$ is, it should have the property that it is preserved under rotations and reversed by reflections. For example, in $\mathbb{R}^2$ the notions of 'clockwise' and 'counterclockwise' certainly have this property, as do 'right-handed' and 'left-handed' in $\mathbb{R}^3$. We shall take the viewpoint that this property is what characterizes orientations, so anything satisfying the property can be regarded as an orientation.

With this in mind, we propose the following as an algebraic-topological definition: An orientation of $\mathbb{R}^n$ at a point x is a choice of generator of the infinite cyclic group $H_n(\mathbb{R}^n, \mathbb{R}^n - \{x\})$, where the absence of a coefficient group from the notation means that we take coefficients in $\mathbb{Z}$. To verify that the characteristic property of orientations is satisfied we use the isomorphisms $H_n(\mathbb{R}^n, \mathbb{R}^n - \{x\}) \approx H_{n-1}(\mathbb{R}^n - \{x\}) \approx H_{n-1}(S^{n-1})$ where S^{n-1} is a sphere centered at x. Since these isomorphisms are natural, and rotations of S^{n-1} have degree 1, being homotopic to the identity, while reflections have degree -1, we see that a rotation ρ of $\mathbb{R}^n$ fixing x takes a generator α of $H_n(\mathbb{R}^n, \mathbb{R}^n - \{x\})$ to itself, $\rho_*(\alpha) = \alpha$, while a reflection takes α to $-\alpha$.

Note that with this definition, an orientation of $\mathbb{R}^n$ at a point x determines an orientation at every other point y via the canonical isomorphisms $H_n(\mathbb{R}^n, \mathbb{R}^n - \{x\}) \approx H_n(\mathbb{R}^n, \mathbb{R}^n - B) \approx H_n(\mathbb{R}^n, \mathbb{R}^n - \{y\})$ where B is any ball containing both x and y.

An advantage of this definition of local orientation is that it can be applied to any n-dimensional manifold M: A **local orientation** of M at a point x is a choice of generator μ_x of the infinite cyclic group $H_n(M, M - \{x\})$.

Notational Convention. In what follows we will very often be looking at homology groups of the form $H_n(X, X - A)$. To simplify notation we will write $H_n(X, X - A)$ as $H_n(X|A)$, or more generally $H_n(X|A;G)$ if a coefficient group G needs to be specified. By excision, $H_n(X|A)$ depends only on a neighborhood of the closure of A in X, so it makes sense to view $H_n(X|A)$ as *local homology of X at A*.

Having settled what local orientations at points of a manifold are, a global orientation ought to be 'a consistent choice of local orientations at all points.' We make this precise by the following definition. An **orientation** of an n-dimensional manifold M is a function $x \mapsto \mu_x$ assigning to each $x \in M$ a local orientation $\mu_x \in H_n(M|x)$, satisfying the 'local consistency' condition that each $x \in M$ has a neighborhood $\mathbb{R}^n \subset M$ containing an open ball B of finite radius about x such that all the local orientations μ_y at points $y \in B$ are the images of one generator μ_B of $H_n(M|B) \approx H_n(\mathbb{R}^n|B)$ under the natural maps $H_n(M|B) \to H_n(M|y)$. If an orientation exists for M, then M is called **orientable**.

Every manifold M has an orientable two-sheeted covering space $\widetilde{M}$. For example, $\mathbb{R}P^2$ is covered by S^2, and the Klein bottle has the torus as a two-sheeted covering space. The general construction goes as follows. As a set, let

$$\widetilde{M} = \{\, \mu_x \mid x \in M \text{ and } \mu_x \text{ is a local orientation of } M \text{ at } x \,\}$$

The map $\mu_x \mapsto x$ defines a two-to-one surjection $\widetilde{M} \to M$, and we wish to topologize $\widetilde{M}$ to make this a covering space projection. Given an open ball $B \subset \mathbb{R}^n \subset M$ of finite radius and a generator $\mu_B \in H_n(M|B)$, let $U(\mu_B)$ be the set of all $\mu_x \in \widetilde{M}$ such that $x \in B$ and μ_x is the image of μ_B under the natural map $H_n(M|B) \to H_n(M|x)$. It is easy to check that these sets $U(\mu_B)$ form a basis for a topology on $\widetilde{M}$, and that the projection $\widetilde{M} \to M$ is a covering space. The manifold $\widetilde{M}$ is orientable since each point $\mu_x \in \widetilde{M}$ has a canonical local orientation given by the element $\widetilde{\mu}_x \in H_n(\widetilde{M}|\mu_x)$ corresponding to μ_x under the isomorphisms $H_n(\widetilde{M}|\mu_x) \approx H_n(U(\mu_B)|\mu_x) \approx H_n(B|x)$, and by construction these local orientations satisfy the local consistency condition necessary to define a global orientation.

Proposition 3.25. *If M is connected, then M is orientable iff $\widetilde{M}$ has two components. In particular, M is orientable if it is simply-connected, or more generally if $\pi_1(M)$ has no subgroup of index two.*

The first statement is a formulation of the intuitive notion of nonorientability as being able to go around some closed loop and come back with the opposite orientation, since in terms of the covering space $\widetilde{M} \to M$ this corresponds to a loop in M that lifts to a path in $\widetilde{M}$ connecting two distinct points with the same image in M. The existence of such paths is equivalent to $\widetilde{M}$ being connected.

Proof: If M is connected, $\widetilde{M}$ has either one or two components since it is a two-sheeted covering space of M. If it has two components, they are each mapped homeomorphically to M by the covering projection, so M is orientable, being homeomorphic to a component of the orientable manifold $\widetilde{M}$. Conversely, if M is orientable, it has exactly two orientations since it is connected, and each of these orientations defines a component of $\widetilde{M}$. The last statement of the proposition follows since connected two-sheeted covering spaces of M correspond to index-two subgroups of $\pi_1(M)$, by the classification of covering spaces. $\square$

The covering space $\widetilde{M} \to M$ can be embedded in a larger covering space $M_{\mathbb{Z}} \to M$ where $M_{\mathbb{Z}}$ consists of all elements $\alpha_x \in H_n(M|x)$ as x ranges over M. As before, we topologize $M_{\mathbb{Z}}$ via the basis of sets $U(\alpha_B)$ consisting of α_x's with $x \in B$ and α_x the image of an element $\alpha_B \in H_n(M|B)$ under the map $H_n(M|B) \to H_n(M|x)$. The covering space $M_{\mathbb{Z}} \to M$ is infinite-sheeted since for fixed $x \in M$, the α_x's range over the infinite cyclic group $H_n(M|x)$. Restricting α_x to be zero, we get a copy M_0 of M in $M_{\mathbb{Z}}$. The rest of $M_{\mathbb{Z}}$ consists of an infinite sequence of copies M_k of $\widetilde{M}$, $k = 1, 2, \cdots$, where M_k consists of the α_x's that are k times either generator of $H_n(M|x)$.

A continuous map $M \to M_{\mathbb{Z}}$ of the form $x \mapsto \alpha_x \in H_n(M|x)$ is called a **section** of the covering space. An orientation of M is the same thing as a section $x \mapsto \mu_x$ such that μ_x is a generator of $H_n(M|x)$ for each x.

One can generalize the definition of orientation by replacing the coefficient group $\mathbb{Z}$ by any commutative ring R with identity. Then an ***R*-orientation** of M assigns to each $x \in M$ a generator of $H_n(M|x;R) \approx R$, subject to the corresponding local consistency condition, where a 'generator' of R is an element u such that $Ru = R$. Since we assume R has an identity element, this is equivalent to saying that u is a unit, an invertible element of R. The definition of the covering space $M_{\mathbb{Z}}$ generalizes immediately to a covering space $M_R \to M$, and an R-orientation is a section of this covering space whose value at each $x \in M$ is a generator of $H_n(M|x;R)$.

The structure of M_R is easy to describe. In view of the canonical isomorphism $H_n(M|x;R) \approx H_n(M|x) \otimes R$, each $r \in R$ determines a subcovering space M_r of M_R consisting of the points $\pm\mu_x \otimes r \in H_n(M|x;R)$ for μ_x a generator of $H_n(M|x)$. If r has order 2 in R then $r = -r$ so M_r is just a copy of M, and otherwise M_r is isomorphic to the two-sheeted cover $\widetilde{M}$. The covering space M_R is the union of these M_r's, which are disjoint except for the equality $M_r = M_{-r}$.

In particular we see that an orientable manifold is R-orientable for all R, while a nonorientable manifold is R-orientable iff R contains a unit of order 2, which is equivalent to having $2 = 0$ in R. Thus every manifold is $\mathbb{Z}_2$-orientable. In practice this means that the two most important cases are $R = \mathbb{Z}$ and $R = \mathbb{Z}_2$. In what follows the reader should keep these two cases foremost in mind, but we will usually state results for a general R.

The orientability of a closed manifold is reflected in the structure of its homology, according to the following result.

Theorem 3.26. *Let M be a closed connected n-manifold. Then*:

(a) *If M is R-orientable, the map $H_n(M;R) \to H_n(M\,|\,x;R) \approx R$ is an isomorphism for all $x \in M$.*

(b) *If M is not R-orientable, the map $H_n(M;R) \to H_n(M\,|\,x;R) \approx R$ is injective with image $\{\, r \in R \mid 2r = 0 \,\}$ for all $x \in M$.*

(c) *$H_i(M;R) = 0$ for $i > n$.*

In particular, $H_n(M;\mathbb{Z})$ is $\mathbb{Z}$ or 0 depending on whether M is orientable or not, and in either case $H_n(M;\mathbb{Z}_2) = \mathbb{Z}_2$.

An element of $H_n(M;R)$ whose image in $H_n(M\,|\,x;R)$ is a generator for all x is called a **fundamental class** for M with coefficients in R. By the theorem, a fundamental class exists if M is closed and R-orientable. To show that the converse is also true, let $\mu \in H_n(M;R)$ be a fundamental class and let μ_x denote its image in $H_n(M\,|\,x;R)$. The function $x \mapsto \mu_x$ is then an R-orientation since the map $H_n(M;R) \to H_n(M\,|\,x;R)$ factors through $H_n(M\,|\,B;R)$ for B any open ball in M containing x. Furthermore, M must be compact since μ_x can only be nonzero for x in the image of a cycle representing μ, and this image is compact. In view of these remarks a fundamental class could also be called an **orientation class** for M.

The theorem will follow fairly easily from a more technical statement:

Lemma 3.27. *Let M be a manifold of dimension n and let $A \subset M$ be a compact subset. Then*:

(a) *If $x \mapsto \alpha_x$ is a section of the covering space $M_R \to M$, then there is a unique class $\alpha_A \in H_n(M\,|\,A;R)$ whose image in $H_n(M\,|\,x;R)$ is α_x for all $x \in A$.*

(b) *$H_i(M\,|\,A;R) = 0$ for $i > n$.*

To deduce the theorem from this, choose $A = M$, a compact set by assumption. Part (c) of the theorem is immediate from (b) of the lemma. To obtain (a) and (b) of the theorem, let $\Gamma_R(M)$ be the set of sections of $M_R \to M$. The sum of two sections is a section, and a scalar multiple of a section is a section, so $\Gamma_R(M)$ is an R-module. There is a homomorphism $H_n(M;R) \to \Gamma_R(M)$ sending a class α to the section $x \mapsto \alpha_x$, where α_x is the image of α under the map $H_n(M;R) \to H_n(M\,|\,x;R)$. Part (a) of the lemma asserts that this homomorphism is an isomorphism. If M is connected, each section is uniquely determined by its value at one point, so statements (a) and (b) of the theorem are apparent from the earlier discussion of the structure of M_R. $\square$

Proof of 3.27: The coefficient ring R will play no special role in the argument so we shall omit it from the notation. We break the proof up into four steps.

(1) First we observe that if the lemma is true for compact sets A, B, and $A \cap B$, then it is true for $A \cup B$. To see this, consider the Mayer–Vietoris sequence

$$0 \longrightarrow H_n(M \mid A \cup B) \xrightarrow{\Phi} H_n(M \mid A) \oplus H_n(M \mid B) \xrightarrow{\Psi} H_n(M \mid A \cap B)$$

Here the zero on the left comes from the assumption that $H_{n+1}(M \mid A \cap B) = 0$. The map Φ is $\Phi(\alpha) = (\alpha, -\alpha)$ and Ψ is $\Psi(\alpha, \beta) = \alpha + \beta$, where we omit notation for maps on homology induced by inclusion. The terms $H_i(M \mid A \cup B)$ farther to the left in this sequence are sandwiched between groups that are zero by assumption, so $H_i(M \mid A \cup B) = 0$ for $i > n$. This gives (b). For the existence half of (a), if $x \mapsto \alpha_x$ is a section, the hypothesis gives unique classes $\alpha_A \in H_n(M \mid A)$, $\alpha_B \in H_n(M \mid B)$, and $\alpha_{A \cap B} \in H_n(M \mid A \cap B)$ having image α_x for all x in A, B, or $A \cap B$ respectively. The images of α_A and α_B in $H_n(M \mid A \cap B)$ satisfy the defining property of $\alpha_{A \cap B}$, hence must equal $\alpha_{A \cap B}$. Exactness of the sequence then implies that $(\alpha_A, -\alpha_B) = \Phi(\alpha_{A \cup B})$ for some $\alpha_{A \cup B} \in H_n(M \mid A \cup B)$. This means that $\alpha_{A \cup B}$ maps to α_A and α_B, so $\alpha_{A \cup B}$ has image α_x for all $x \in A \cup B$ since α_A and α_B have this property. To see that $\alpha_{A \cup B}$ is unique, observe that if a class $\alpha \in H_n(M \mid A \cup B)$ has image zero in $H_n(M \mid x)$ for all $x \in A \cup B$, then its images in $H_n(M \mid A)$ and $H_n(M \mid B)$ have the same property, hence are zero by hypothesis, so α itself must be zero since Φ is injective. Uniqueness of $\alpha_{A \cup B}$ follows by applying this observation to the difference between two choices for $\alpha_{A \cup B}$.

(2) Next we reduce to the case $M = \mathbb{R}^n$. A compact set $A \subset M$ can be written as the union of finitely many compact sets $A_1, \cdots, A_m$ each contained in an open $\mathbb{R}^n \subset M$. We apply the result in (1) to $A_1 \cup \cdots \cup A_{m-1}$ and A_m. The intersection of these two sets is $(A_1 \cap A_m) \cup \cdots \cup (A_{m-1} \cap A_m)$, a union of $m-1$ compact sets each contained in an open $\mathbb{R}^n \subset M$. By induction on m this gives a reduction to the case $m = 1$. When $m = 1$, excision allows us to replace M by the neighborhood $\mathbb{R}^n \subset M$.

(3) When $M = \mathbb{R}^n$ and A is a union of convex compact sets $A_1, \cdots, A_m$, an inductive argument as in (2) reduces to the case that A itself is convex. When A is convex the result is evident since the map $H_i(\mathbb{R}^n \mid A) \rightarrow H_i(\mathbb{R}^n \mid x)$ is an isomorphism for any $x \in A$, as both $\mathbb{R}^n - A$ and $\mathbb{R}^n - \{x\}$ deformation retract onto a sphere centered at x.

(4) For an arbitrary compact set $A \subset \mathbb{R}^n$ let $\alpha \in H_i(\mathbb{R}^n \mid A)$ be represented by a relative cycle z, and let $C \subset \mathbb{R}^n - A$ be the union of the images of the singular simplices in ∂z. Since C is compact, it has a positive distance δ from A. We can cover A by finitely many closed balls of radius less than δ centered at points of A. Let K be the union of these balls, so K is disjoint from C. The relative cycle z defines an element $\alpha_K \in H_i(\mathbb{R}^n \mid K)$ mapping to the given $\alpha \in H_i(\mathbb{R}^n \mid A)$. If $i > n$ then by (3) we have $H_i(\mathbb{R}^n \mid K) = 0$, so $\alpha_K = 0$, which implies $\alpha = 0$ and hence $H_i(\mathbb{R}^n \mid A) = 0$. If $i = n$ and α_x is zero in $H_n(\mathbb{R}^n \mid x)$ for all $x \in A$, then in fact this holds for all $x \in K$, where α_x in this case means the image of α_K. This is because K is a union of balls B meeting A and $H_n(\mathbb{R}^n \mid B) \rightarrow H_n(\mathbb{R}^n \mid x)$ is an isomorphism for all $x \in B$. Since

$\alpha_x = 0$ for all $x \in K$, (3) then says that α_K is zero, hence also α. This finishes the uniqueness statement in (a). The existence statement is easy since we can let α_A be the image of the element α_B associated to any ball $B \supset A$. $\square$

For a closed n-manifold having the structure of a Δ-complex there is a more explicit construction for a fundamental class. Consider the case of $\mathbb{Z}$ coefficients. In simplicial homology a fundamental class must be represented by some linear combination $\sum_i k_i\sigma_i$ of the n-simplices σ_i of M. The condition that the fundamental class maps to a generator of $H_n(M\,|\,x;\mathbb{Z})$ for points x in the interiors of the σ_i's means that each coefficient k_i must be ± 1. The k_i's must also be such that $\sum_i k_i\sigma_i$ is a cycle. This implies that if σ_i and σ_j share a common $(n-1)$-dimensional face, then k_i determines k_j and vice versa. Analyzing the situation more closely, one can show that a choice of signs for the k_i's making $\sum_i k_i\sigma_i$ a cycle is possible iff M is orientable, and if such a choice is possible, then the cycle $\sum_i k_i\sigma_i$ defines a fundamental class. With $\mathbb{Z}_2$ coefficients there is no issue of signs, and $\sum_i \sigma_i$ always defines a fundamental class.

Some information about $H_{n-1}(M)$ can also be squeezed out of the preceding theorem:

Corollary 3.28. *If M is a closed connected n-manifold, the torsion subgroup of $H_{n-1}(M;\mathbb{Z})$ is trivial if M is orientable and $\mathbb{Z}_2$ if M is nonorientable.*

Proof: This is an application of the universal coefficient theorem for homology, using the fact that the homology groups of M are finitely generated, from Corollaries A.8 and A.9 in the Appendix. In the orientable case, if $H_{n-1}(M;\mathbb{Z})$ contained torsion, then for some prime p, $H_n(M;\mathbb{Z}_p)$ would be larger than the $\mathbb{Z}_p$ coming from $H_n(M;\mathbb{Z})$. In the nonorientable case, $H_n(M;\mathbb{Z}_m)$ is either $\mathbb{Z}_2$ or 0 depending on whether m is even or odd. This forces the torsion subgroup of $H_{n-1}(M;\mathbb{Z})$ to be $\mathbb{Z}_2$. $\square$

The reader who is familiar with Bockstein homomorphisms, which are discussed in §3.E, will recognize that the $\mathbb{Z}_2$ in $H_{n-1}(M;\mathbb{Z})$ in the nonorientable case is the image of the Bockstein homomorphism $H_n(M;\mathbb{Z}_2)\to H_{n-1}(M;\mathbb{Z})$ coming from the short exact sequence of coefficient groups $0\to\mathbb{Z}\to\mathbb{Z}\to\mathbb{Z}_2\to 0$.

The structure of $H_n(M;G)$ and $H_{n-1}(M;G)$ for a closed connected n-manifold M can be explained very nicely in terms of cellular homology when M has a CW structure with a single n-cell, which is the case for a large number of manifolds. Note that there can be no cells of higher dimension since a cell of maximal dimension produces nontrivial local homology in that dimension. Consider the cellular boundary map $d: C_n(M)\to C_{n-1}(M)$ with $\mathbb{Z}$ coefficients. Since M has a single n-cell we have $C_n(M) = \mathbb{Z}$. If M is orientable, d must be zero since $H_n(M;\mathbb{Z}) = \mathbb{Z}$. Then since d is zero, $H_{n-1}(M;\mathbb{Z})$ must be free. On the other hand, if M is nonorientable then d

must take a generator of $C_n(M)$ to twice a generator α of a $\mathbb{Z}$ summand of $C_{n-1}(M)$, in order for $H_n(M;\mathbb{Z}_p)$ to be zero for odd primes p and $\mathbb{Z}_2$ for $p=2$. The cellular chain α must be a cycle since 2α is a boundary and hence a cycle. It follows that the torsion subgroup of $H_{n-1}(M;\mathbb{Z})$ must be a $\mathbb{Z}_2$ generated by α.

Concerning the homology of noncompact manifolds there is the following general statement.

Proposition 3.29. *If M is a connected noncompact n-manifold, then $H_i(M;R)=0$ for $i\geq n$.*

Proof: Represent an element of $H_i(M;R)$ by a cycle z. This has compact image in M, so there is an open set $U\subset M$ containing the image of z and having compact closure $\overline{U}\subset M$. Let $V=M-\overline{U}$. Part of the long exact sequence of the triple $(M,U\cup V,V)$ fits into a commutative diagram

$$\begin{tikzcd}
H_{i+1}(M,U\cup V;R) \arrow[r] & H_i(U\cup V,V;R) \arrow[r] & H_i(M,V;R) \\
& H_i(U;R) \arrow[r] \arrow[u, "\approx"'] & H_i(M;R) \arrow[u]
\end{tikzcd}$$

When $i>n$, the two groups on either side of $H_i(U\cup V,V;R)$ are zero by Lemma 3.27 since $U\cup V$ and V are the complements of compact sets in M. Hence $H_i(U;R)=0$, so z is a boundary in U and therefore in M, and we conclude that $H_i(M;R)=0$.

When $i=n$, the class $[z]\in H_n(M;R)$ defines a section $x\mapsto[z]_x$ of M_R. Since M is connected, this section is determined by its value at a single point, so $[z]_x$ will be zero for all x if it is zero for some x, which it must be since z has compact image and M is noncompact. By Lemma 3.27, z then represents zero in $H_n(M,V;R)$, hence also in $H_n(U;R)$ since the first term in the upper row of the diagram above is zero when $i=n$, by Lemma 3.27 again. So $[z]=0$ in $H_n(M;R)$, and therefore $H_n(M;R)=0$ since $[z]$ was an arbitrary element of this group. $\square$

The Duality Theorem

The form of Poincaré duality we will prove asserts that for an R-orientable closed n-manifold, a certain naturally defined map $H^k(M;R)\to H_{n-k}(M;R)$ is an isomorphism. The definition of this map will be in terms of a more general construction called *cap product*, which has close connections with cup product.

For an arbitrary space X and coefficient ring R, define an R-bilinear cap product $\frown : C_k(X;R)\times C^\ell(X;R)\to C_{k-\ell}(X;R)$ for $k\geq\ell$ by setting

$$\sigma\frown\varphi=\varphi(\sigma\,|\,[v_0,\cdots,v_\ell])\,\sigma\,|\,[v_\ell,\cdots,v_k]$$

for $\sigma:\Delta^k\to X$ and $\varphi\in C^\ell(X;R)$. To see that this induces a cap product in homology